

INFORMATION THEORY & CODING

Review Summary

■ McMillan inequality

Uniquely decodable codes $\Leftrightarrow \sum D^{-\ell_i} \leq 1$.

■ Huffman code

$$L^* = \min_{\sum D^{-\ell_i} \leq 1} \sum p_i \ell_i$$

$$H_D(X) \leq L^* < H_D(X) + 1.$$

Coin tossing vs. Poker

- Toss a fair coin and see the sequences

Head, Tail, Tail, Head, ...

$$p(x_1, x_2, \dots, x_n) \approx 2^{-nH(X)}$$

- Play card games and see the sequence



$$p(x_1, x_2, \dots, x_n) = ?$$

Recall AEP

- **AEP Theorem** “Almost all events are almost equally surprising.” Specifically, if $X_1, X_2, \dots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X) \text{ in probability.}$$

On average, $nH(X)$ bits suffice to describe n i.i.d. random variables. But what if the random variables are *dependent*? In particular, what if the random variables form a *stationary process*?

How to Model dependence: Markov Chains

- A *stochastic process* $\{X_i\}$ is an indexed sequence of random variables $(X_0, X_1, \dots)$ characterized by the joint PMF $\Pr[(X_1, X_2, \dots, X_n) = (x_1, x_2, \dots, x_n)] = p(x_1, x_2, \dots, x_n)$, where $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$ for $n = 0, 1, \dots$

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- **Definition** A *stochastic process* is said to be *stationary* (稳态的) if the joint distribution of any subset of the sequence of random variables is **invariant** with respect to shifts in the time index, i.e.,

$$\begin{aligned} & \Pr[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \\ &= \Pr[X_{1+\ell} = x_1, X_{2+\ell} = x_2, \dots, X_{n+\ell} = x_n] \end{aligned}$$

for every n and every shift ℓ and for all

5-2 $x_1, x_2, \dots, x_n \in \mathcal{X}$.

Markov Chains

- **Definition** A discrete stochastic process $X_1, X_2, \dots$ is said to be a *Markov chain* or a *Markov process* if for $n = 1, 2, \dots$,

$$\begin{aligned}\Pr[X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_1 = x_1] \\ = \Pr[X_{n+1} = x_{n+1} | X_n = x_n]\end{aligned}$$

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for all $x_1, x_2, \dots, x_n, x_{n+1} \in \mathcal{X}$.

In this case, the joint PMF(联合概率质量函数) can be written as

$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_2) \cdots p(x_n|x_{n-1}).$$

Markov Chains

- **Definition** The Markov chain is called *time invariant* (时间不变性) if the conditional probability $p(x_{n+1}|x_n)$ does **NOT** depend on n , i.e., for $n = 1, 2, \dots$,

$$\Pr[X_{n+1} = b | X_n = a] = \Pr[X_2 = b | X_1 = a] \quad \text{for all } a, b \in \mathcal{X}.$$

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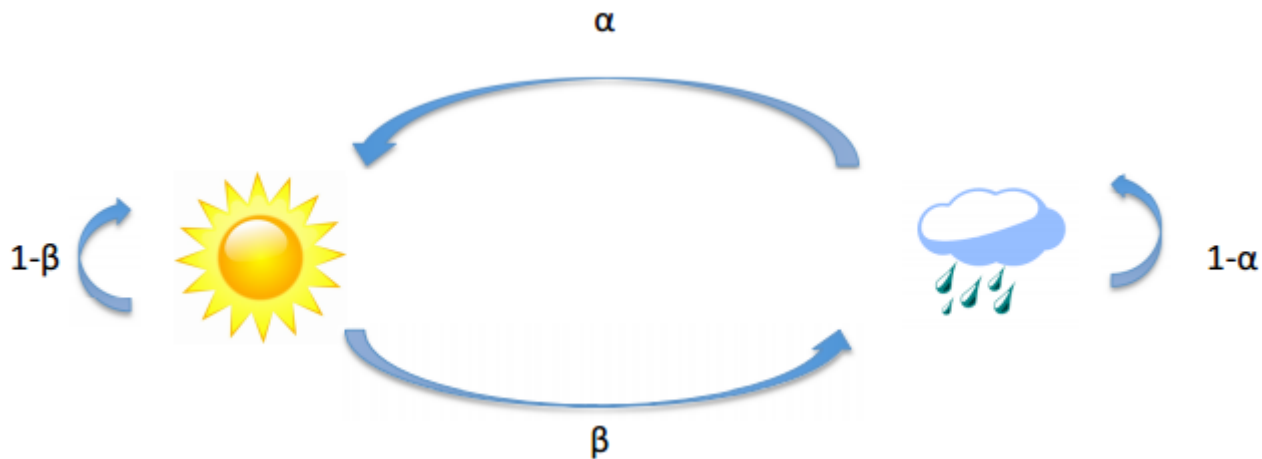
We deal with *time invariant Markov chains*. If $\{X_i\}$ is a Markov chain, X_n is called the *state* at time n . A time invariant Markov chain is characterized by its initial state and a *probability transition matrix* $P = [P_{ij}]$, $i, j \in \{1, 2, \dots, m\}$, where $P_{ij} = \Pr[X_{n+1} = j | X_n = i]$.

Simple weather model

■ $\mathcal{X} = \{\text{Sunny: } S, \text{ Rainy: } R\}$

$$p(S|S) = 1 - \beta, \quad p(R|R) = 1 - \alpha, \quad p(R|S) = \beta, \quad p(S|R) = \alpha$$

$$P = \begin{bmatrix} 1 - \beta & \beta \\ \alpha & 1 - \alpha \end{bmatrix}$$



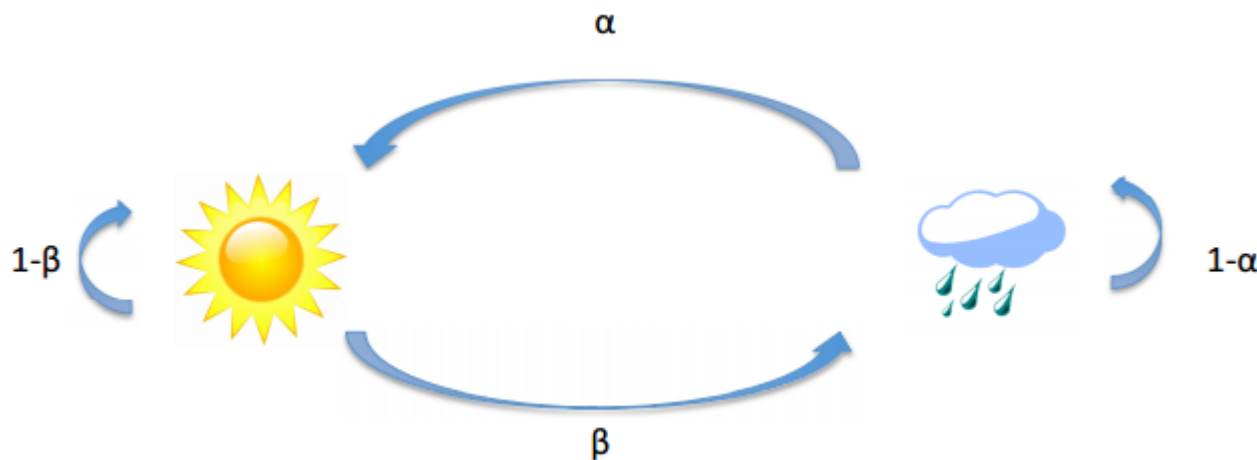
Simple weather model

- Probability of seeing a sequence SSRR:

$$p(SSRR) = p(S)p(S|S)p(R|S)p(R|R) = p(S)(1 - \beta)\beta(1 - \alpha)$$

What will this sequence behave, after many days of observations?

- What sequences of observations are more *typical*?
- What is the probability of seeing a typical sequence?



Stationary Distribution

- If the PMF of the random variable at time n is $p(x_n)$, the PMF at time $n + 1$ is

$$p(x_{n+1}) = \sum_{x_n} p(x_n) P_{x_n x_{n+1}}.$$

A distribution μ on the states such that the distribution at time $n + 1$ is the same as the distribution at time n is called *a stationary distribution*.

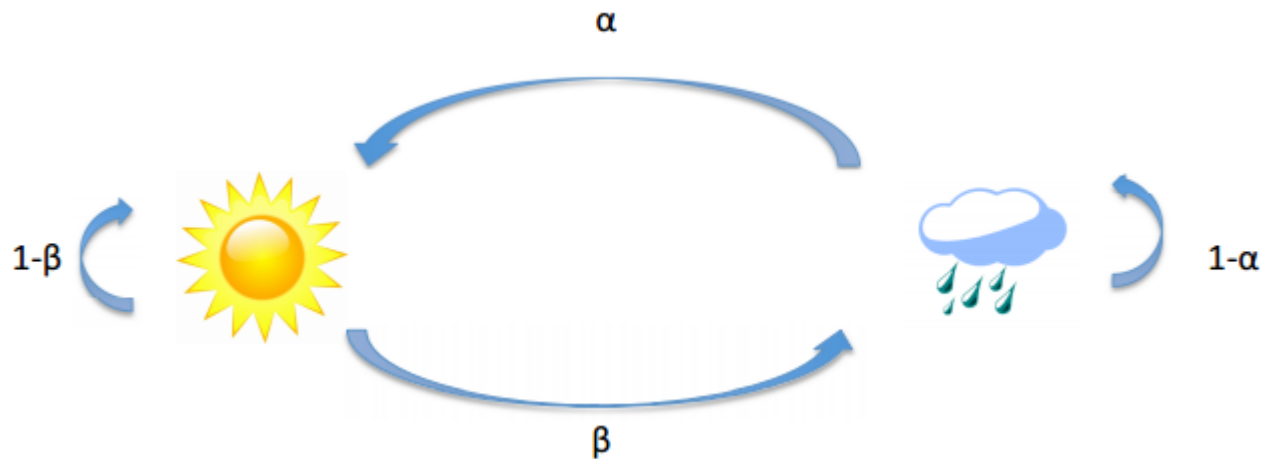
Stationary Distribution

- – If $\mu(S) = \frac{\alpha}{\alpha+\beta}$, $\mu(R) = \frac{\beta}{\alpha+\beta}$

$$P = \begin{bmatrix} 1-\beta & \beta \\ \alpha & 1-\alpha \end{bmatrix}$$

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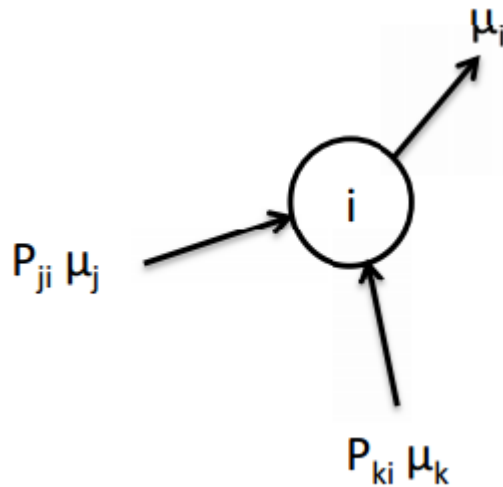
$$\begin{aligned} p(X_{n+1} = S) &= p(S|S)\mu(S) + p(S|R)\mu(R) \\ &= (1-\beta)\frac{\alpha}{\alpha+\beta} + \alpha\frac{\beta}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta} = \mu(S). \end{aligned}$$



Stationary Distribution

- How to calculate stationary distribution?
 - Stationary distribution $\mu_i, i = 1, 2, \dots, |\mathcal{X}|$ satisfies

$$\mu_i = \sum_j \mu_j p_{ji} \quad (\mu = \mu P), \text{ and } \sum_{i=1}^{|\mathcal{X}|} \mu_i = 1.$$



Entropy Rate

- When X_i 's are i.i.d., the entropy

$$H(X^n) = H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i) = nH(X).$$

Entropy Rate

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- With **dependent** sequences X_i 's, how does $H(X^n)$ grow with n ?

Entropy Rate

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- With **dependent** sequences X_i 's, how does $H(X^n)$ grow with n ?
- *Entropy rate* characterized the growth rate.

Entropy Rate

- **Definition 1:** average entropy per symbol

$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} \frac{H(X_1, X_2, \dots, X_n)}{n}$$

- **Definition 2:** conditional entropy of the last r.v. given the past

$$H'(\mathcal{X}) = \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, X_{n-2}, \dots, X_1)$$

Entropy Rate

- **Theorem 4.2.2** For a stationary stochastic process, $H(X_n|X_{n-1}, \dots, X_1)$ is nonincreasing in n and has a limit $H'(\mathcal{X})$.

Entropy Rate

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Proof.

$$\begin{aligned} H(X_{n+1}|X_1, X_2, \dots, X_n) &\stackrel{\text{conditioning reduces entropy}}{\leq} H(X_{n+1}|X_n, \dots, X_2) \\ &\stackrel{\text{stationarity}}{=} H(X_n|X_{n-1}, \dots, X_1), \end{aligned}$$

- $H(X_n|X_{n-1}, \dots, X_1)$ decreases as n increases
- $H(X) \geq 0$
- The limit must exist.

Entropy Rate

- **Theorem 4.2.1** For a stationary stochastic process,
 $H(\mathcal{X}) = H'(\mathcal{X})$.

Entropy Rate

- **Theorem 4.2.1** For a stationary stochastic process, $H(\mathcal{X}) = H'(\mathcal{X})$.

Proof. By the chain rule,

$$\frac{1}{n}H(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$$

- $H(X_n | X_{n-1}, \dots, X_1) \rightarrow H'(\mathcal{X})$
- *Cesaro mean*:
If $a_n \rightarrow a$, $b_n = \frac{1}{n} \sum_{i=1}^n a_i$, then $b_n \rightarrow a$.

- So

$$\frac{1}{n}H(X_1, \dots, X_n) \rightarrow H'(\mathcal{X})$$

AEP for stationary ergodic process (chap 16)

$$-\frac{1}{n} \log p(X_1, \dots, X_n) \rightarrow H(\mathcal{X})$$

- $p(X_1, \dots, X_n) \approx 2^{-nH(\mathcal{X})}$
- Typical sequences in typical set of size $2^{-nH(\mathcal{X})}$
- We can use $nH(\mathcal{X})$ bits to represent typical sequences

Entropy Rate for Markov Chain

- For a stationary Markov chain, the entropy rate is

$$\begin{aligned} H(\mathcal{X}) &= H'(\mathcal{X}) = \lim H(X_n | X_{n-1}, \dots, X_1) = \lim H(X_n | X_{n-1}) \\ &= H(X_2 | X_1). \end{aligned}$$

- By definition

$$p(X_2 = j | X_1 = i) = P_{ij}$$

- Entropy rate of stationary Markov chain

$$H(\mathcal{X}) = H(X_2 | X_1) = \sum_i \mu_i \left(\sum_j -P_{ij} \log P_{ij} \right) = - \sum_{ij} \mu_i P_{ij} \log P_{ij}.$$

To Calculate Entropy Rate

- 1. Find *stationary distribution* μ_i

$$\mu_i = \sum_j \mu_j p_{ji} \quad (\mu = \mu P), \text{ and } \sum_{i=1}^{|\mathcal{X}|} \mu_i = 1.$$

- 2. Use *transition probability* P_{ij}

$$H(\mathcal{X}) = - \sum_{ij} \mu_i P_{ij} \log P_{ij}$$

Entropy Rate of Weather Model

- Stationary distribution $\mu(S) = \frac{\alpha}{\alpha+\beta}$, $\mu(R) = \frac{\beta}{\alpha+\beta}$

$$P = \begin{bmatrix} 1 - \beta & \beta \\ \alpha & 1 - \alpha \end{bmatrix}$$

$$\begin{aligned} H(\mathcal{X}) &= \mu(S)H(\beta) + \mu(R)H(\alpha) \\ &= \frac{\alpha}{\alpha + \beta}H(\beta) + \frac{\beta}{\alpha + \beta}H(\alpha) \\ &\stackrel{\text{Jensen's inequality}}{\leq} H\left(2\frac{\alpha\beta}{\alpha + \beta}\right) \end{aligned}$$

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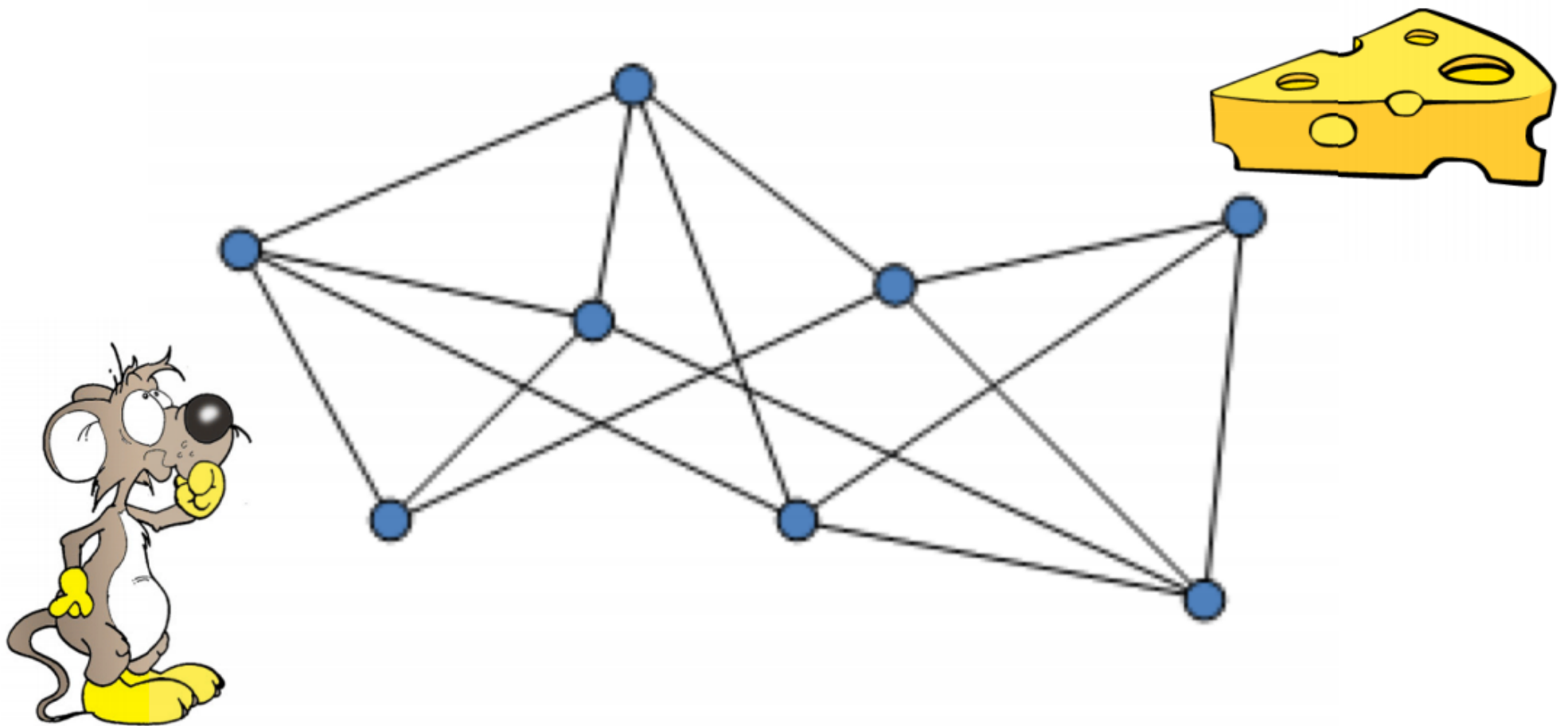
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Maximum when $\alpha = \beta = 1/2$: degenerate to independent process

Random Walk on a Weighted Graph

- An **undirected graph** with m nodes $\{1, 2, \dots, m\}$
- Edge $i \rightarrow j$ has weight $W_{ij} \geq 0$ ($W_{ij} = W_{ji}$)
- A particle walks **randomly** from node to node
- Random walk $X_1, X_2, \dots$: a sequence of vertices
- Given $X_n = i$, next step chosen from neighboring nodes with probability

$$P_{ij} = \frac{W_{ij}}{\sum_k W_{ik}}$$



Entropy Rate of Random Walk on a Graph

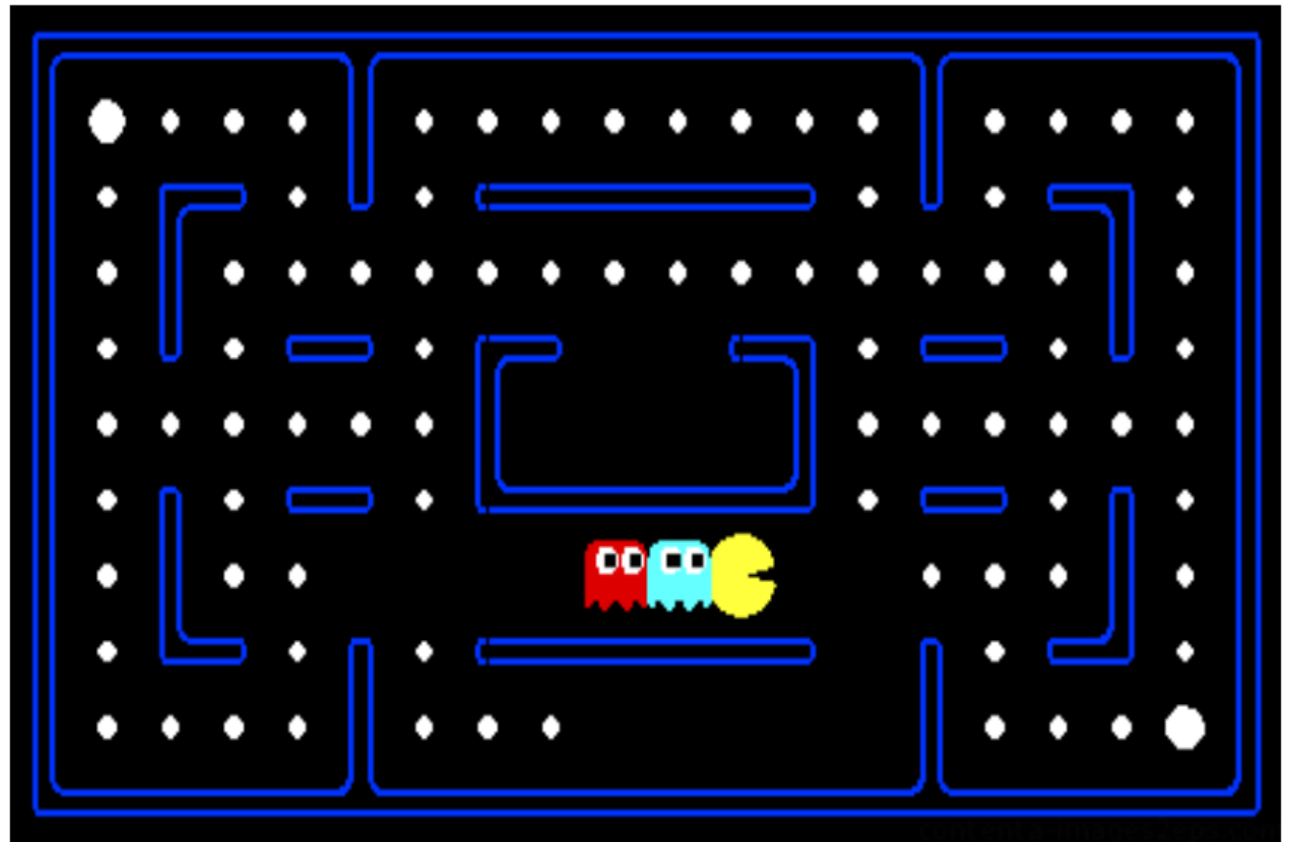
- Let

$$W_i = \sum_j W_{ij}, \quad W = \sum_{i,j: j>i} W_{ij}$$

- The stationary distribution is

$$\mu_i = \frac{W_i}{2W}$$

- Can verify this is a stationary distribution by checking $\mu P = \mu$
- The stationary probability of state i is **proportional** to the weight of edges emanating from node i (**locality**)
- The entropy rate can thus be calculated



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